Session Objectives

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At the end of this session, you will be able to

1.Explain the pathogenesis of **acute tubular injury**

2. Become familiar with the mechanisms leading to **tubulointerstitial nephritis**

3. and various forms of **pyelonephritis and tubulointerstitial nephritis induced by drugs** including their pathogenesis and morphological features

4. Acquire a good understanding of **urate nephropathy,** its pathogenesis and histological features.

Please note that unless mentioned otherwise, all the images and information included in this presentation are from Robbins and Cotran, Pathologic Basis of Disease, 10th edition

Acute Tubular Injury/ Acute Tubular Necrosis (ATI/ATN)

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ATI =**tubular injury** (necrosis of tubular epithelial cells) and **acute renal failure**
Necrosis may be absent. Therefore, the term ATI is preferred over the older name acute tubular necrosis (ATN).

ATI can be caused by a variety of conditions, including the following:

Ischemia=due to decreased or interrupted blood flow :

- diffuse involvement of the intrarenal blood vessels such as in microscopic polyangiitis, microangiopathies (e.g., hemolytic uremic syndrome [HUS]
- thrombotic thrombocytopenic purpura [TTP])
- decreased effective circulating blood volume, as occurs in hypovolemic shock

Direct toxic injury to the tubules

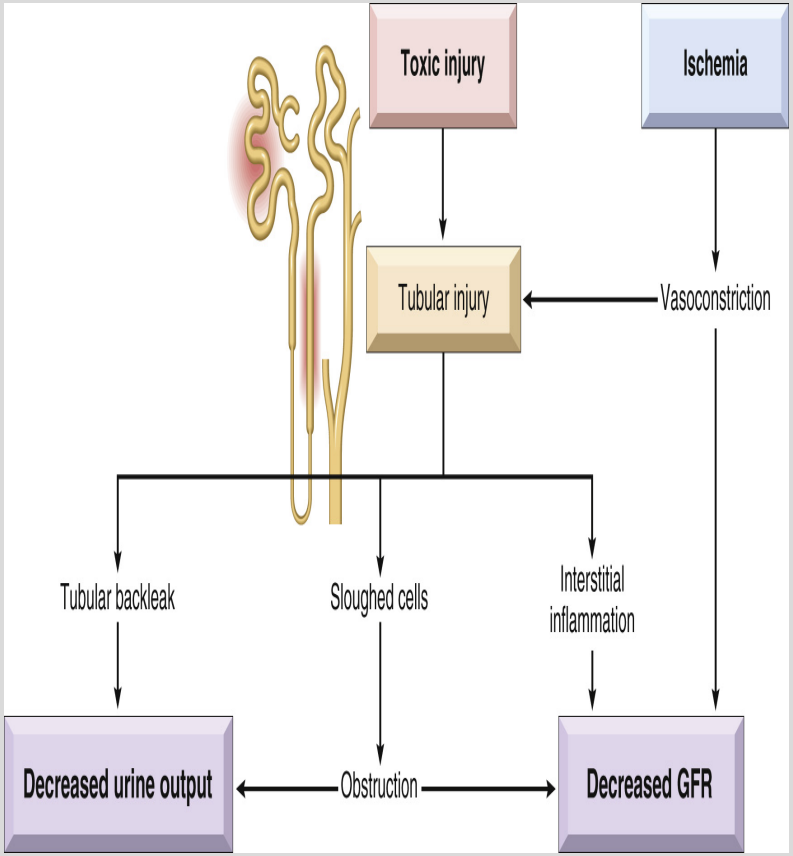
- endogenous agents: myoglobin, hemoglobin, monoclonal light chains, bile/bilirubin
- exogenous agents: drugs, radiocontrast dyes, heavy metals, organic solvents

Pathogenesis of ATI

Disturbance in blood flow and Tubular cell injury

Tubular epithelial cells (proximal tubules): are sensitive to ischemia and toxins.

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Ischemia causes numerous structural and functional alterations in epithelial cells:

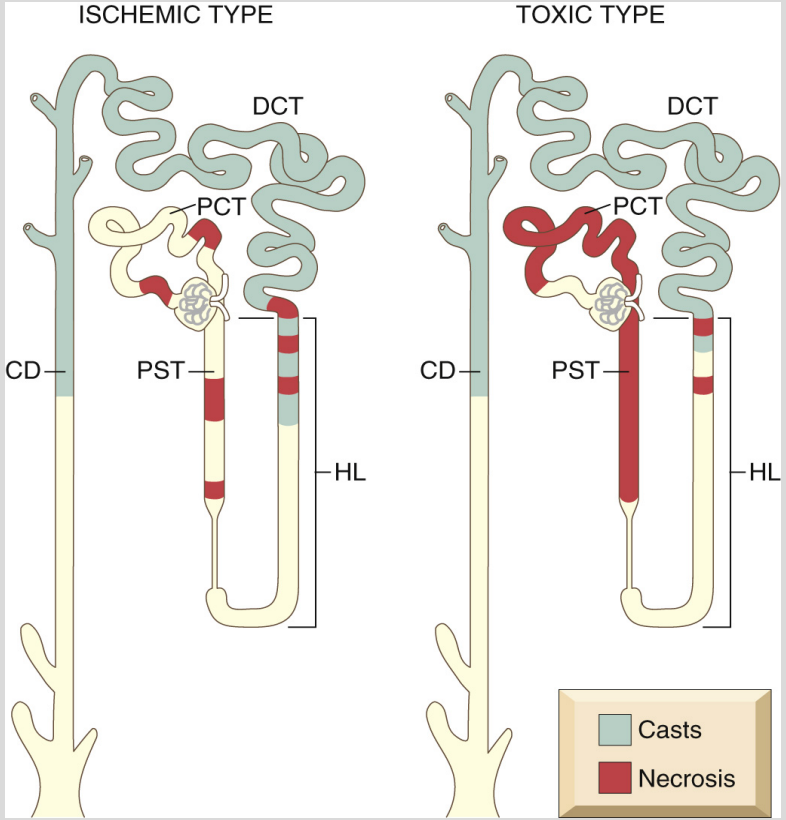
- Loss of cell polarity due to redistribution of membrane proteins such the enzyme Na,K + -ATPase)
- It results in an abnormal ion transport across the cells and increased sodium supply to distal tubules, which produces vasoconstriction.

Subsequently damaged tubular cells induces production of cytokines and leukocytes are attracted promoting further injury.

Injured cells detach from the basement membranes causing luminal obstruction; this increases the intratubular pressure, and further decrease in GFR.

ATI Morphology

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Patterns of tubular damage

In the ischemic type:
-tubular necrosis is patchy, short lengths of tubules are affected
-proximal straight tubule (PST) segments
-ascending limbs of Henle's loop (HL)/ rarely proximal convoluted tubule (PCT).

In toxic acute tubular type:
-extensive necrosis is present along the proximal convoluted tubule (PCT
but necrosis of the distal tubule, particularly ascending HL (HL) is seen.

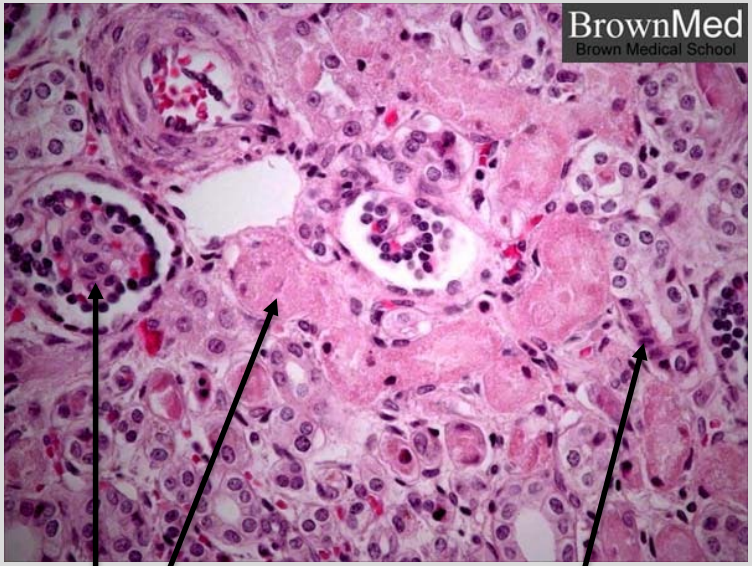
In both types, lumens of the distal convoluted tubules (DCT) and collecting ducts (CD) contain casts.

ATI = tubular epithelial injury at multiple points along the nephron, with large skip areas in between,
-+- rupture of basement membranes
-+- occlusion of tubular lumens by casts.

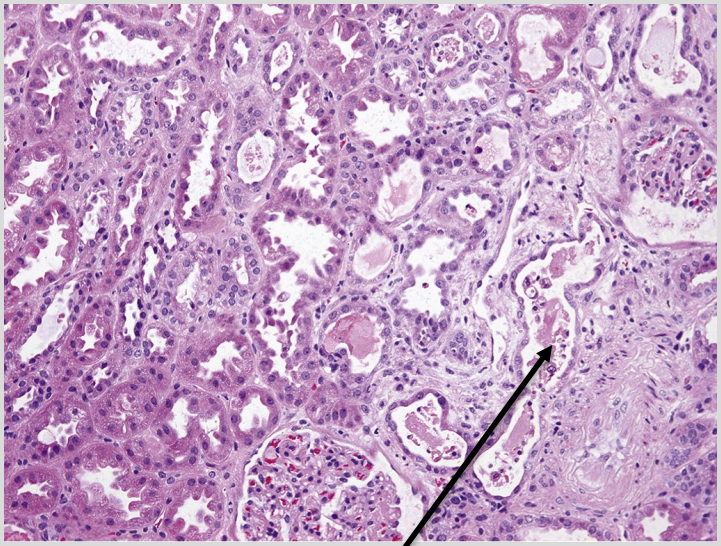
ATI Histology

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Necrosis and sloughing of epithelial cells of the proximal convoluted tubules. The glomeruli and distal convoluted tubules are preserved.



Some tubular epithelial cells in the tubules are necrotic, and many have detached (from their basement membranes) and been sloughed into the tubular lumens.

Tubulointerstitial Nephritis

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This group of renal diseases is characterized by inflammatory injuries of the tubules and interstitium.

Tubulointerstitial disorders are distinguished clinically from the glomerular diseases by the following:

- Absence of nephritic or nephrotic syndrome
- Presence of defects in tubular function

-impaired ability to concentrate urine:
polyuria or nocturia; salt wasting
-diminished ability to excrete acids (metabolic acidosis)
- isolated defects in tubular reabsorption or secretion.

Tubulointerstitial nephritis can be acute or chronic.

Acute tubulointerstitial nephritis
-rapid clinical onset
-interstitial edema, leukocytic infiltration of the interstitium and tubules, and tubular injury.

Chronic interstitial nephritis
-infiltration with predominantly mononuclear leukocytes,
-interstitial fibrosis, tubular atrophy

Primary tubulointerstitial nephritis

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Infections
Acute bacterial pyelonephritis
Chronic pyelonephritis (including reflux nephropathy)
Other infections (e.g., viruses, parasites)
Toxins
Drugs
Acute-hypersensitivity interstitial nephritis
Analgesics
Heavy metals
Lead, cadmium

Metabolic Diseases
Urate nephropathy
Nephrocalcinosis (hypercalcemic nephropathy)
Oxalate nephropathy
Physical Factors
Chronic urinary tract obstruction
Neoplasms
Multiple myeloma (light-chain cast nephropathy)
Immunologic Reactions
Transplant rejection
Sjögren syndrome
Sarcoidosis

Vascular Diseases
Miscellaneous
Nephronophthisis
“Idiopathic” interstitial nephritis

Secondary tubulointerstitial nephritis

Polycystic kidney disease

Diabetes

It may also contribute to progressive damage.

Pyelonephritis

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=inflammation affecting the tubules, interstitium, and renal pelvis.

Acute pyelonephritis :caused by bacterial infection and is associated with urinary tract infection.

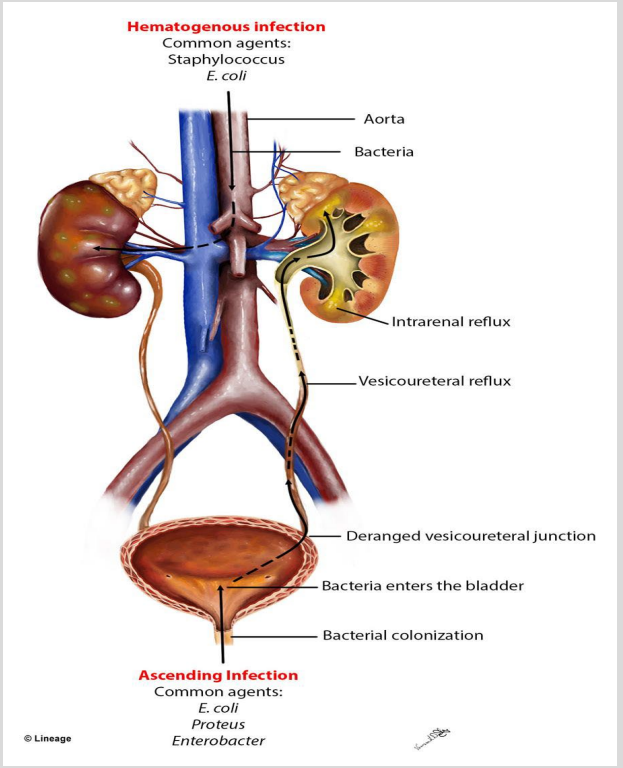
Chronic pyelonephritis: more complex disorder; bacterial infection plays a dominant role, but there are other predisposing factors such as: vesicoureteral reflux, obstruction

Mechanisms

Hematogenous

- seeding of the kidneys by bacteria from distant foci in the course of septicemia
- localized infections such as infective endocarditis.
- ureteral obstruction
- immunosuppressed patients.

Ascending: combination of urinary bladder infection, vesicoureteral reflux, and intrarenal reflux



Acute Pyelonephritis

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It is a suppurative inflammation of the kidney caused by bacterial and sometimes viral (polyomavirus) infection.

Causes : ascending UTI (E coli) or a hematogenous spread.
The patients present with WBC in the urine and/or WBC casts.

Clinical: fever, flank pain, nausea, vomiting, chills.

Complications: development of chronic pyelonephritis

Papillary necrosis: in diabetics, sickle cell disease, and in those with urinary tract obstruction. One or all of the pyramids of the affected kidney may be involved.

Pyonephrosis :total or almost complete obstruction. The suppurative exudate is unable to drain and fills the renal pelvis, calyces, and ureter with pus.

Perinephric abscess: extension of suppurative inflammation through the renal capsule into the perinephric tissue.

Chronic Pyelonephritis

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It is a chronic tubulointerstitial inflammation and scarring involve the calyces and pelvis

It results from a recurrent or an inadequately treated acute pyelonephritis in the presence of VUR or chronically obstructing renal stones.
It is seen in 10% to 20% of patients in renal transplant or dialysis units.

Clinical : silent gradual onset of renal insufficiency and hypertension. Later they develop proteinuria and end-stage renal failure

Tubulointerstitial Nephritis Induced by Drugs

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Drugs and toxic agents represent the second most common cause of primary tubulointerstitial nephritis (after pyelonephritis).

Mechanism
Immunologic reaction: drug haptene binds tubules triggering an IG E reaction to haptene bound tubular cells and their basement membrane.

Drugs: sulphonamides, methicillin, ampicillin, rifampin, diuretics, cimetidine

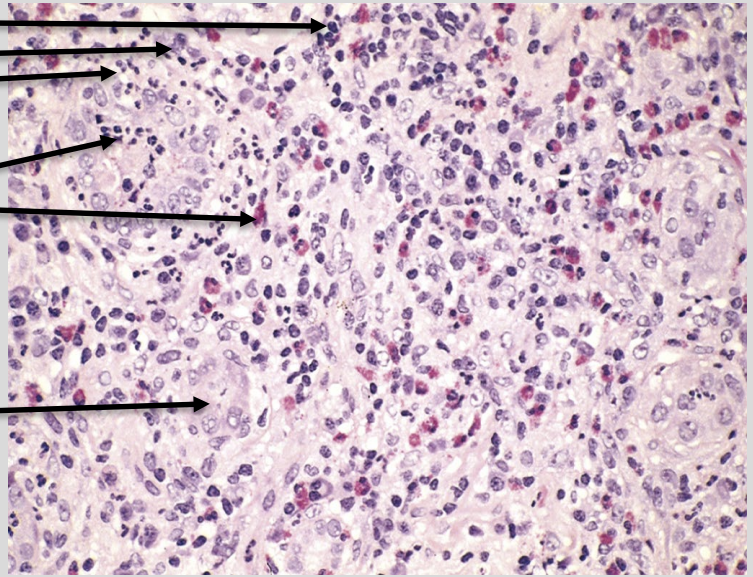
Clinical: 15 days post exposure to drug, fever, eosinophilia, skin rash, oliguria in 50% of cases.

Acute Drug-Induced Interstitial Nephritis: Morphology

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Interstitial edema and infiltration by
Lymphocytes
Macrophages
Neutrophils
Eosinophils

Tubulitis: infiltration of tubules by
PMN
Lymphocytes



NSAIDs Tubulointerstitial Nephritis

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Often NSAIDs are frequently administered to patients at risk for renal disease.
Many NSAIDs are nonselective cyclo-oxygenase inhibitors but also inhibit cyclo-oxygenase–dependent prostaglandin synthesis.
The selective COX-2 inhibitors, while sparing the gastrointestinal tract, do affect the kidneys because COX-2 is expressed in human kidneys.

- **Acute kidney injury**: decreased synthesis of vasodilatory prostaglandins and resultant ischemia.

- **Acute tubulointerstitial nephritis:** hypersensitivity immunological mechanism (as described before)

- **Acute tubulointerstitial nephritis and minimal change disease=** renal failure + nephrotic syndrome

- **Membranous nephropathy**, with the nephrotic syndrome: unclear pathogenesis**.**

Metabolic Tubulointerstitial Diseases: Urate Nephropathy

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Acute uric acid nephropathy

- It those with leukemias or lymphomas after chemotherapy (tumor lysis syndrome)
- The drugs kill tumor cells, and uric acid is produced as released nucleic acids are broken down.
- Precipitation of uric acid is favored by the acidic pH in collecting tubules.
- This is followed by **obstruction of nephrons** and the development of **acute renal failure.**

Chronic urate nephropathy, or gouty nephropathy

It is rare; in protracted forms of hyperuricemia.

- There are monosodium urate crystals deposits in the acidic milieu of the distal tubules and collecting ducts forming **distinct birefringent needle-like crystals** either in the tubular lumens or in the interstitium.
- These urate deposits determine a mononuclear response that contains foreign-body giant cells.
- A tophus may be formed leading to **tubular obstruction** by the urates which causes **cortical atrophy and scarring**.